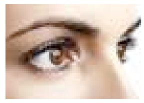
Xenobots
Scientists reassemble a frog's living matter into robotic devices; no electronics required. **http://dgit.in/jan21-05**



LHON Gene Therapy
Injecting an experimental gene therapy vector in one eye surprisingly boosts vision in both. **http://dgit.in/jan21-06**

Programmable Medicine

Advances in biological circuits have opened up the possibility of medicines that react to local conditions within the body

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Credit: Allison Carter, Georgia Tech

They look just like regular drugs

Logic gates can be considered the most basic parts of a computer, and are pathways that allow boolean operations, having one or many inputs and a single output. The operations are based on the binary values of 0 and 1. In an AND gate, only if both the inputs are 1, then the result is 1. In an OR gate, if either of the inputs is 1, then the output is 1. Computers can be built from a series of logic gates, with researchers from Kobe University in Japan even making functional computers using mazes and swarms of crabs. In synthetic biology, it is possible to make logic gates using DNA, which was first demonstrated by researchers from Boston University in 2000. A fairly complex circuit called the repressilator created a negative feedback loop, where genes that coded for three protein components, each repressed one of the other protein components, leading to a perpetual game of rock paper scissors. These genetic circuits depend on components with known interactions, but the development is slow because of the unknown interactions between genes. However, since then, there have been many advancements in making complex DNA machines, including robots, calculators and computers. In fact, there is a citizen science gaming project that gives you the tools needed to make your own DNA machines. The project is called Nanocrafter, and can be accessed on nanocrafter.org.

In 2019, researchers from Harvard used the repressilator as a genetic clock, to measure how bacteria grow over time, by tracking the number of cycles that the repressilator goes through. The researchers found the cycle repeats after 15.5 bacterial generations. The researchers were even able to track the expression of the proteins over time by tagging the proteins with fluorescent markers. The repressilator stayed stable for 16 days within mice guts, which demonstrated its value as a viable monitoring tool. The repressilator is a